Beyond the Average Treatment Effect

The `lmtp` package

MODIFIED
April 18, 2025

So far, we've

1. Defined a causal estimand for general, hypothetical interventions
2. Identified that estimand from observational data
3. Discussed 4 different estimators.

It's now time to move on to applying this new knowledge to real data examples.

Existing software and `lmtp`

General software for estimation of causal effects in longitudinal studies is limited:

- The `ipw` and `gfoRmula` packages provide routines for estimating causal effects using inverse probability weighting (IPW) and the parametric g-formula respectively. As we already discussed, however, the validity of these methods requires making unrealistic parametric assumptions.
- The `ltmle` and `survtmle` packages implement a doubly-robust method for estimating causal effects from longitudinal data, but these packages do not support continuous valued exposures, multiple exposures, interventions that depend on the natural value of the exposure, or stochastic interventions.

In contrast, the `lmtp` package (software companion to the paper by Díaz et al. (2023)) can be used to doubly robustly and flexibly estimate point-in-time and longitudinal effects of the general set of interventions that we've talked about today.

- `lmtp` could be your default package for conducting causal analyses in R!

Installation

For this workshop, `lmtp` has already been installed with `webr`. However, you can install the package locally on your machine from CRAN with

```
install.packages("lmtp")
```

Alternatively, you can download the developmental version from GitHub with

```
# install.packages(devtools)
devtools::install_github("nt-williams/lmtp@devel")
```

Preliminaries

Before we move on to using `lmtp`, here's general information that will be applicable across all examples:

- Data is passed to estimators through the `data` argument.
- Data should be in wide format with one column per variable per time point under study (i.e., there should be one column for every variable in O). Data may be either a `data.frame` or `tibble` but not a `data.table`.
- Columns do not have to be in any specific order and the data may contain variables that are not used in estimation.
- The names of treatment variables are specified with the `trt` argument.
- The names of censoring variables are specified with the `cens` argument.
- The names of baseline covariates are specified with the `baseline` argument.
- The names of time-varying covariates are specified with the `time_vary` argument.
- The `trt` argument accepts either a character vector or a list of character vectors.
- The `cens` and `baseline` arguments accept character vectors.
- The `time_vary` argument accepts a list of character vectors.

> Vectors and lists are basic data structures in R. A vector can be thought of as a 1-dimensional collection of homogeneous elements. For example a character vector is a collection of values with the class `character`.
>
> ```
> a_character_vector <- c("this", "is", "a", "character", "vector")
> ```
>
> A list is similar, except that it can hold elements of different types.
>
> ```
> a_list <- list(
>  a_character_vector = c("this", "is", "a", "character", "vector"),
>  a_numeric_vector = 1:10,
> ```

```
  a_list_whithin_a_list = list(1, 2, 3, 4)
)
```

- The `trt`, `cens`, and `time_vary` arguments must be sorted according to the time-ordering of the model with each index containing the name (or names) of variables for the given time.
- The outcome variable is specified with the `outcome` argument.
- The `outcome_type` argument specifies the type of outcome. It should be set to `"continuous"` for continuous outcomes, `"binomial"` for dichotomous outcomes, and `"survival"` for time-to–event outcomes.
- Censoring indicators should be coded using 0 and 1 where 1 indicates an observation is observed at the next time and 0 indicates loss-to-follow-up. Once an observation's censoring status is switched to 0 it cannot change back to 1. Missing data before an observation is censored is not allowed.
- The argument should be set to for continuous outcomes, for dichotomous outcomes, and for time-to-event outcomes.

Data structure examples

Point treatment

Baseline confounders

Trt. Outcome

ID	W1	W2	W3	W4	A	Y
1	50	0	1012	23	0	0
2	68	1	25300	17	0	1
3	32	1	255	95	1	0
...	...	...	...	...	...	...
N	79	0	6783	13	0	NA

Adapted from Hoffman et al., 2022.

Point-treatment with censoring

Baseline confounders

Trt.

Censoring

Outcome

ID	W1	W2	W3	W4	A	C	Y
1	50	0	1012	23	0	1	0
2	68	1	25300	17	0	1	1
3	32	1	255	95	1	1	0
...	...	...	...	...	...	...	...
N	79	0	6783	13	0	0	NA

Adapted from Hoffman et al., 2022.

Time-varying treatment with censoring

Baseline confounders

Time-dependent confounders

Trt.

Censoring

Outcome

ID	W1	W2	$L1_1$	$L2_2$	A_1	C_1	$L1_2$	$L2_2$	A_2	C_2	...	Y_{T+1}
1	50	0	1012	23	0	1	1901	23	0	1	...	1
2	68	1	25300	17	0	1	NA	NA	NA	NA	...	1
3	32	1	255	95	1	1	270	52	1	1	...	0
...	...	...	...	...	...	...	...	...	...	...	...	...
N	79	0	6783	13	0	0	NA	NA	NA	NA	...	NA

First time point *Second time point*

Adapted from Hoffman et al., 2022.

Point-treatment with survival outcome

Baseline confounders | Time-dependent confounders | Trt. | Censoring | Survival

ID	W1	W2	$L1_1$	$L2_2$	A	C_1	Y_2	C_2	Y_3	...	Y_{T+1}
1	50	0	1012	23	0	1	0	1	0	...	1
2	68	1	25300	17	0	1	1	NA	1	...	1
3	32	1	255	95	1	1	0	1	0	...	0
...	...	...	...	...	...	...	...	...	...	...	...
N	79	0	6783	13	0	0	NA	NA	NA	...	NA

Adapted from Hoffman et al., 2022.

Time-varying treatment with survival outcome

Baseline confounders | Time-dependent confounders | Trt. | Censoring | Survival

ID	W1	W2	$L1_1$	$L2_2$	A_1	C_1	Y_2	$L1_2$	$L2_2$	A_2	C_2	Y_3	...	Y_{T+1}
1	50	0	1012	23	0	1	0	1901	23	0	1	0	...	1
2	68	1	25300	17	0	1	1	NA	NA	NA	NA	1	...	1
3	32	1	255	95	1	1	0	270	52	1	1	0	...	0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
N	79	0	6783	13	0	0	NA	NA	NA	NA	NA	NA	...	NA

First time point | *Second time point*

Adapted from Hoffman et al., 2022.

Machine learning ensembles

As was already discussed, an attractive property of multiply-robust estimators is that they can incorporate flexible machine-learning algorithms for the estimation of nuisance parameters while remaining $\sqrt{n}$-consistent.

- `lmtp` uses the super learner algorithm for estimating these nuisance parameters.
- The super learner algorithm is an ensemble learner than incorporates a set of candidate models through a weighted convex-combination based on cross-validation. Asymptotically, this weighted combination of models, called the meta-learner, will outperform any single one of its components.

- `lmtp` uses the implementation of the super learner provided by the `SuperLearner` package.
- The algorithms to be used in the super learner are specified with the `learners_trt` and `learners_outcome` arguments.
- The outcome variable type should guide users on selecting the appropriate candidate learners for use with the `learners_outcome` argument.
- Regardless of whether an exposure is continuous, dichotomous, or categorical, the exposure mechanism is estimated using classification. Therefore only include candidate learners capable of binary classification with the `learners_trt` argument.
- Candidate learners that rely on cross-validation for the tuning of hyper-parameters should support grouped data if used with `learners_trt`. Because estimation of the treatment mechanism relies on the augmented $2n$ duplicated data set, duplicated observations must be put into the same fold during sample-splitting. This is done automatically by the package.

References

Díaz, Iván, Nicholas Williams, Katherine L Hoffman, and Edward J Schenck. 2023. "Nonparametric Causal Effects Based on Longitudinal Modified Treatment Policies." *Journal of the American Statistical Association* 118 (542): 846–57.

Van der Laan, Mark J, Eric C Polley, and Alan E Hubbard. 2007. "Super Learner." *Statistical Applications in Genetics and Molecular Biology* 6 (1).

Williams, Nicholas, and Iván Díaz. 2023. "Lmtp: An R Package for Estimating the Causal Effects of Modified Treatment Policies." *Observational Studies* 9 (2): 103–22.